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📘 STUDIFY — Complete System Documentation

> **Student Task Management System** Generated: March 12, 2026 | Schema Version: v3.1

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1. System Overview

Property	Value
Stack	PHP 8+, MySQL/MariaDB, Bootstrap 5.3, Vanilla JS
Server	Laragon (Apache) at `localhost/Studify/`
Database	MySQL `studify`, charset `utf8mb4`
Auth	Session-based with bcrypt password hashing

Property	Value
Theme	Green primary (`#16A34A`), white sidebar, dark mode toggle
PWA	Standalone display, manifest + icons
Architecture	Server-rendered PHP with AJAX enhancements
CSS	Single `style.css` with CSS custom properties for theming
JS	Single `main.js` (1053 lines), modular object pattern, zero jQuery

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2. Database Schema

2.1 `users`

Column	Type	Notes
`id`	INT PK AUTO_INCREMENT	Primary key
`name`	VARCHAR(255)	NOT NULL
`email`	VARCHAR(255) UNIQUE	NOT NULL, indexed
`password`	VARCHAR(255)	bcrypt hash
`role`	ENUM('student','admin')	Default `student`, indexed
`course`	VARCHAR(255)	e.g. "BS Information Systems"
`year_level`	INT	1–6
`profile_photo`	VARCHAR(500)	Relative path to uploaded avatar
`onboarding_completed`	TINYINT(1)	Default 0
`login_attempts`	INT	Brute-force counter, default 0
`locked_until`	DATETIME	Account lockout expiry
`created_at`	TIMESTAMP	Auto-set
`updated_at`	TIMESTAMP	ON UPDATE auto-set

2.2 `password_resets`

Column	Type	Notes
`id`	INT PK AI	
`email`	VARCHAR(255)	Indexed
`token`	VARCHAR(255)	SHA-256 hash of raw token, indexed
`expires_at`	DATETIME	1-hour expiry
`used`	TINYINT(1)	Default 0
`created_at`	TIMESTAMP	

2.3 `semesters`

Column	Type	Notes
`id`	INT PK AI	
`user_id`	INT FK → users	ON DELETE CASCADE
`name`	VARCHAR(255)	e.g. "1st Semester 2025-2026"
`is_active`	BOOLEAN	Default 0, indexed
`created_at` / `updated_at`	TIMESTAMP	

2.4 `subjects`

Column	Type	Notes
`id`	INT PK AI	
`semester_id`	INT FK → semesters	ON DELETE CASCADE
`name`	VARCHAR(255)	
`instructor_name`	VARCHAR(255)	
`created_at` / `updated_at`	TIMESTAMP	

2.5 `tasks`

Column	Type	Notes
id	INT PK AI	
user_id	INT FK → users	ON DELETE CASCADE
subject_id	INT FK → subjects	ON DELETE SET NULL, nullable
parent_id	INT FK → tasks (self)	ON DELETE CASCADE — for subtasks
title	VARCHAR(255)	
description	TEXT	
deadline	DATETIME	Indexed
priority	ENUM('Low','Medium','High')	Default 'Medium'
type	ENUM('Assignment','Quiz','Project','Exam','Report','Other')	Default 'Assignment'
status	ENUM('Pending','In Progress','Completed')	Default 'Pending', indexed
is_recurring	TINYINT(1)	Default 0
recurrence_type	ENUM('Daily','Weekly','Monthly')	
recurrence_end	DATE	
position	INT	Reserved for future task ordering
created_at / updated_at	TIMESTAMP	

2.6 study_sessions

Column	Type	Notes
id	INT PK AI	
user_id	INT FK → users	ON DELETE CASCADE
task_id	INT FK → tasks	ON DELETE SET NULL, nullable
subject_id	INT FK → subjects	ON DELETE SET NULL, nullable
duration	INT	Minutes
session_type	ENUM('Focus','Break')	Default 'Focus'
created_at	TIMESTAMP	Indexed

2.7 notes

Column	Type	Notes
id	INT PK AI	
subject_id	INT FK → subjects	ON DELETE SET NULL, nullable
user_id	INT FK → users	ON DELETE CASCADE
title	VARCHAR(255)	
content	LONGTEXT	Rich HTML (Quill.js) or plain text
content_type	ENUM('plain','markdown')	Default 'markdown'
created_at / updated_at	TIMESTAMP	

2.8 attachments

Column	Type	Notes
id	INT PK AI	
user_id	INT FK → users	ON DELETE CASCADE

Column	Type	Notes
`task_id`	INT FK → tasks	ON DELETE CASCADE, nullable
`note_id`	INT FK → notes	ON DELETE CASCADE, nullable
`file_name`	VARCHAR(255)	Original filename
`file_path`	VARCHAR(500)	Relative storage path
`file_size`	INT	Bytes
`file_type`	VARCHAR(100)	MIME type
`created_at`	TIMESTAMP	

2.9 `announcements`

Column	Type	Notes
`id`	INT PK AI	
`admin_id`	INT FK → users	ON DELETE CASCADE
`title`	VARCHAR(255)	
`content`	TEXT	
`priority`	ENUM('Low','Normal','Important','Urgent')	Default 'Normal'
`expires_at`	DATE	Nullable
`created_at` / `updated_at`	TIMESTAMP	

2.10 `announcement_reads`

Column	Type	Notes
`id`	INT PK AI	
`announcement_id`	INT FK → announcements	ON DELETE CASCADE
`user_id`	INT FK → users	ON DELETE CASCADE
`read_at`	TIMESTAMP	
	UNIQUE KEY	`(announcement_id, user_id)`

2.11 study_buddies

Column	Type	Notes
id	INT PK AI	
requester_id	INT FK → users	ON DELETE CASCADE
partner_id	INT FK → users	ON DELETE CASCADE
status	ENUM('pending','accepted','declined','unlinked')	Default 'pending'
invite_code	VARCHAR(32)	Unique 8-char hex code
created_at / updated_at	TIMESTAMP	
	UNIQUE KEY	(requester_id, partner_id)

2.12 buddy_nudges

Column	Type	Notes
id	INT PK AI	
sender_id / receiver_id	INT FK → users	ON DELETE CASCADE
message	VARCHAR(500)	
is_read	TINYINT(1)	Default 0
created_at	TIMESTAMP	

2.13 buddy_messages

Column	Type	Notes
id	INT PK AI	
sender_id / receiver_id	INT FK → users	ON DELETE CASCADE

Column	Type	Notes
`message`	TEXT	
`message_type`	ENUM('text','nudge','emoji','system')	Default 'text'
`reply_to_id`	INT FK → buddy_messages	ON DELETE SET NULL — reply threading
`is_read`	TINYINT(1)	Default 0
`created_at`	TIMESTAMP	Composite indexes for conversations

2.14 `buddy_typing_status`

Column	Type	Notes
`user_id`	INT PK FK → users	ON DELETE CASCADE
`typing_until`	TIMESTAMP	3-second TTL

2.15 `buddy_blocks`

Column	Type	Notes
`id`	INT PK AI	
`blocker_id` / `blocked_id`	INT FK → users	ON DELETE CASCADE
`created_at`	TIMESTAMP	
	UNIQUE KEY	`(blocker_id, blocked_id)`

2.16 `buddy_reports`

Column	Type	Notes
`id`	INT PK AI	
`reporter_id` / `reported_id`	INT FK → users	ON DELETE CASCADE
`reason`	ENUM('harassment','spam','inappropriate','impersonation','other')	

Column	Type	Notes
`details`	TEXT	
`status`	ENUM('pending','reviewed','resolved','dismissed')	Default 'pending'
`reviewed_by`	INT FK → users	ON DELETE SET NULL
`reviewed_at`	TIMESTAMP	
`created_at`	TIMESTAMP	

2.17 `activity_log`

Column	Type	Notes
`id`	INT PK AI	
`user_id`	INT FK → users	ON DELETE SET NULL
`action`	VARCHAR(255)	
`details`	TEXT	
`ip_address`	VARCHAR(45)	IPv4/IPv6
`created_at`	TIMESTAMP	

---

3. Configuration & Security

File: `config/db.php`

Connection

- **Driver:** `mysqli` to `localhost`
- **Database:** `studify`
- **Charset:** `utf8mb4`
- **User:** `root` (no password — local dev)

Session Settings

- `use_strict_mode` = 1
- `cookie_httponly` = 1
- `cookie_samesite` = "Lax"

Constants

Constant	Value	Purpose
`MAX_LOGIN_ATTEMPTS`	5	Before lockout
`LOCKOUT_DURATION`	15	Minutes
`MAX_FILE_SIZE`	10,485,760	10 MB
`UPLOAD_DIR`	`uploads/`	Relative to project root
`ALLOWED_FILE_TYPES`	pdf, doc, docx, ppt, pptx, xls, xlsx, txt, jpg, jpeg, png, gif, zip, rar	Whitelist

Utility Functions in `db.php`

Function	Purpose
`cleanInput($data)`	`trim()` + `stripslashes()` + `htmlspecialchars()`
`sanitize($data)`	Alias for `cleanInput()`
`generateCSRFToken()`	Creates `$_SESSION['csrf_token']` via `random_bytes(32)`
`csrfTokenField()` / `getCSRFField()`	Returns `<input type="hidden">` HTML
`validateCSRFToken($token)`	`hash_equals()` comparison
`requireCSRF()`	Validates token, **rotates** (prevents replay), returns 403 for AJAX
`redirect($location, $message, $type)`	Sets flash message in session, `header()` redirect

4. Authentication System

File: `includes/auth.php`

Function	Signature	Purpose
`isLoggedIn()`	`(): bool`	Checks `$_SESSION['user_id']` exists
`isStudent()`	`(): bool`	`$_SESSION['role'] === 'student'`
`isAdminRole()`	`(): bool`	`$_SESSION['role'] === 'admin'`
`requireLogin()`	`(): void`	Redirects to login with `?redirect=` if not logged in; 2-hour idle timeout via `$_SESSION['last_activity']`
`requireAdmin()`	`(): void`	Redirects to index if not admin
`getCurrentUserId()`	`(): ?int`	Returns session user ID
`getCurrentUserRole()`	`(): ?string`	Returns session role
`regenerateSession()`	`(): void`	`session_regenerate_id(true)`
`secureLogin($user)`	`(array): void`	Sets all session vars, regenerates session ID
`secureLogout()`	`(): void`	Full session destruction + cookie clearing
`isAccountLocked($email, $conn)`	`(string, mysqli): bool`	Checks `locked_until`; auto-resets if expired
`incrementLoginAttempts($email, $conn)`	`(string, mysqli): void`	Increments counter; locks at MAX_LOGIN_ATTEMPTS
`resetLoginAttempts($email, $conn)`	`(string, mysqli): void`	Resets on successful login

5. Core PHP Functions

File: `includes/functions.php` (939 lines)

5.1 User & Semester

Function	Parameters	Returns	Purpose
`getUserInfo`	`($user_id, $conn)`	`?array`	Full user row
`getActiveSemester`	`($user_id, $conn)`	`?array`	Active semester (`is_active=1`)
`getUserSemesters`	`($user_id, $conn)`	`array`	All semesters, newest first
`getSemesterSubjects`	`($semester_id, $conn)`	`array`	Subjects in a semester
`isAdmin`	`($user_id, $conn)`	`bool`	DB-level admin check

5.2 Tasks

Function	Parameters	Returns	Purpose
`getSubjectTasks`	`($subject_id, $conn)`	`array`	Parent tasks for a subject
`getUserTasks`	`($user_id, $conn, $limit, $offset)`	`array`	Paginated tasks with joins
`getUserTasksFiltered`	`($user_id, $conn, $subject_id, $status, $sort, $sort_dir)`	`array`	SQL-filtered/sorted tasks; whitelist-validated sort; `FIELD()` for priority
`getTaskStatusCounts`	`($user_id, $conn)`	`array`	Single-query: `total`, `pending`, `in_progress`, `completed`

Function	Parameters	Returns	Purpose
`getSubjectTaskStats`	`($subject_id, $conn)`	`array`	`total` + `completed` for subject cards
`getPendingTasksCount`	`($user_id, $conn)`	`int`	Count of pending parent tasks
`getCompletedTasksCount`	`($user_id, $conn)`	`int`	Count of completed parent tasks
`getTotalTasksCount`	`($user_id, $conn)`	`int`	Count of all parent tasks
`getUpcomingTasks`	`($user_id, $conn, $days)`	`array`	Incomplete tasks within N days
`getCompletionPercentage`	`($user_id, $conn)`	`int`	`completed / total * 100`
`getTasksAsJSON`	`($user_id, $conn)`	`string`	Calendar-formatted JSON with priority colors

5.3 Subtasks

Function	Parameters	Returns	Purpose
`getSubtasks`	`($parent_id, $conn)`	`array`	Child tasks by position
`getSubtaskProgress`	`($parent_id, $conn)`	`array`	`total` + `completed` subtask counts

5.4 Dashboard & Stats

Function	Parameters	Returns	Purpose
`getDashboardStats`	`($user_id, $conn)`	`array`	Single-query: total/pending/completed/in-progress + weekly study minutes

5.5 Formatting & Display

Function	Parameters	Returns	Purpose
`formatDate`	`($date)`	`string`	Format to `M d, Y`
`formatDateTime`	`($datetime)`	`string`	Format to `M d, Y h:i A`
`getPriorityColor`	`($priority)`	`string`	Bootstrap color class
`getStatusColor`	`($status)`	`string`	Bootstrap color class
`getTypeColor`	`($type)`	`string`	Bootstrap color class

5.6 Admin

Function	Parameters	Returns	Purpose
`getAllUsers`	`($conn)`	`array`	All users, newest first
`getTotalSystemTasks`	`($conn)`	`int`	System-wide task count
`getTotalUsers`	`($conn)`	`int`	System-wide user count

5.7 Pagination

Function	Parameters	Returns	Purpose
`paginate`	`($total, $per_page, $current_page)`	`array`	Offset, has_prev, has_next metadata
`renderPagination`	`($pagination, $base_url)`	`string`	Bootstrap pagination HTML

5.8 File Upload

Function	Parameters	Returns	Purpose
`handleFileUpload`	`($file, $user_id, $conn, $task_id, $note_id)`	`array`	Validates, stores in `uploads/{user_id}/`, inserts record

Function	Parameters	Returns	Purpose
`getAttachments`	`($conn, $task_id, $note_id)`	`array`	Fetch attachments for task or note
`formatFileSize`	`($bytes)`	`string`	Human-readable (KB, MB)
`handleProfilePhotoUpload`	`($file, $user_id, $conn)`	`array`	Avatar upload (JPG/PNG/GIF/WebP, max 2MB)

5.9 Onboarding

Function	Parameters	Returns	Purpose
`needsOnboarding`	`($user_id, $conn)`	`bool`	True if no semesters/subjects/tasks and not dismissed
`dismissOnboarding`	`($user_id, $conn)`	`bool`	Sets `onboarding_completed = 1`

5.10 Global Search

Function	Parameters	Returns	Purpose
`globalSearch`	`($user_id, $conn, $query, $limit)`	`array`	Searches tasks, notes, subjects via `LIKE`

5.11 Study Buddy

Function	Parameters	Returns	Purpose
`getAcceptedBuddy`	`($user_id, $conn)`	`?array`	Active buddy (max 1 allowed)
`getLastBuddyPair`	`($user_id, $conn)`	`?array`	Most recent unlinked buddy
`isBuddyBlocked`	`($user_id, $other_id, $conn)`	`bool`	Bidirectional block check

Function	Parameters	Returns	Purpose
`blockBuddy`	`($blocker, $blocked, $conn)`	`bool`	Unpairs + declines + blocks
`unblockBuddy`	`($blocker, $blocked, $conn)`	`bool`	Removes block record
`reportBuddy`	`($reporter, $reported, $reason, $details, $conn)`	`bool`	Creates admin report
`getBlockedUsers`	`($user_id, $conn)`	`array`	List of blocked users
`checkChatRateLimit`	`($user_id, $conn, $max)`	`bool`	Max 15 messages/minute
`getPendingBuddyRequests`	`($user_id, $conn)`	`array`	Incoming pending requests
`getSentBuddyRequest`	`($user_id, $conn)`	`?array`	Outgoing pending request
`generateBuddyCode`	`()`	`string`	8-char uppercase hex
`getBuddyProgress`	`($buddy_id, $conn)`	`array`	Privacy-safe stats for buddy card
`getUnreadNudgeCount`	`($user_id, $conn)`	`int`	Unread nudge count
`getBuddyNudges`	`($user_id, $conn, $limit)`	`array`	Recent nudges with sender info

5.12 Buddy Chat (Instant Messaging)

Function	Parameters	Returns	Purpose
`getChatMessages`	`($user_id, $buddy_id, $conn, $limit, $before_id)`	`array`	Paginated history with reply threading
`getNewChatMessages`	`($user_id, $buddy_id, $conn,`	`array`	New messages since ID (for

Function	Parameters	Returns	Purpose
	`$after_id)`		polling)
`sendChatMessage`	`($sender, $receiver, $message, $conn, $type, $reply_to)`	`int`	Inserts message, returns ID
`markChatMessagesRead`	`($user_id, $sender_id, $conn)`	`int`	Marks all from sender as read
`getUnreadBuddyMessageCount`	`($user_id, $conn)`	`int`	Total unread messages
`deleteChatMessage`	`($message_id, $user_id, $conn)`	`bool`	Sender-only deletion
`updateUserActivity`	`($user_id, $conn)`	`void`	Updates `last_active` timestamp
`isUserOnline`	`($user_id, $conn)`	`bool`	Active within 2 minutes
`updateTypingStatus`	`($user_id, $conn)`	`void`	3-second typing TTL
`clearTypingStatus`	`($user_id, $conn)`	`void`	Clears typing indicator
`isBuddyTyping`	`($buddy_id, $conn)`	`bool`	Checks typing status

6. Student Modules

6.1 Dashboard (`student/dashboard.php`)

- **Features:** Time-based greeting, 4 stat cards (total/pending/completed/in-progress), completion donut chart, priority distribution chart, weekly study hours bar chart, weekly summary, upcoming deadlines, unread announcements
- **Data:** `getDashboardStats()`, `getUpcomingTasks()`, priority queries, weekly charts

6.2 Tasks (`student/tasks.php`)

- **Features:** Card-based task list, add/edit/delete, status toggle, filter by subject & status, sort by deadline/priority/created/title, recurring task generation (Daily/Weekly/Monthly), subtask support
- **Separated views:** Active tasks (Pending/In Progress) shown in main list; Completed tasks in collapsible section below
- **AJAX Endpoints:**
  - `action=toggle_status` → toggles Pending ↔ Completed
  - `action=delete` → deletes task
- **Form Actions:** `add`, `edit`

6.3 Subjects (`student/subjects.php`)

- **Features:** Subject cards with task completion stats, add/edit/delete, auto-selects active semester, ownership verification
- **Form Actions:** `add`, `edit`, `delete`

6.4 Semesters (`student/semesters.php`)

- **Features:** Semester cards with activation toggle, add/edit/delete/rename, prevents deleting active semester, combo dropdown + custom name input for adding
- **Form Actions:** `add`, `edit`, `activate`, `delete`

6.5 Notes (`student/notes.php`)

- **Features:** Quill.js WYSIWYG rich text editor, notes optionally linked to subjects, grid card view with previews, search & filter, view in modal with formatted HTML rendering
- **Rich Text Toolbar:** Headings, bold/italic/underline/strike, text & background color, ordered/unordered lists, blockquote, code block, alignment, links, images, clear formatting
- **Form Actions:** `add`, `edit`, `delete`

6.6 Calendar (`student/calendar.php`)

- **Features:** FullCalendar 6.1.10, tasks as color-coded events (priority-based), drag-to-reschedule, add task from calendar, click event for details

- **AJAX:** `action=update_deadline` → updates task deadline on drag

6.7 Pomodoro Timer (`student/pomodoro.php`)

- **Features:** SVG circular timer, Focus/Short Break/Long Break modes, configurable durations, presets (Classic 25/5, Deep 50/10, Marathon 90/20), auto-start breaks, link to task/subject, mark task complete after session, session history with pagination, daily goal with celebration, persistent timer state across page loads, keyboard shortcuts
- **Focus Ambiance:** 6 procedural Web Audio sounds (Rain, Fire, Waves, Birds, Café, Wind), volume control, presets (Cozy, Nature, Storm, Study Café, Zen), auto-play/pause with timer
- **AJAX Endpoints:**
  - Default POST → saves study session
  - `action=complete_task` → marks task + subtasks as Completed
  - `action=get_history` → paginated session history

6.8 Study Analytics (`student/study_analytics.php`)

- **Features:** Total/weekly/monthly study stats, average session length, most productive day, study streak, 7-day bar chart, 6-month trend line chart, day-of-week radar chart
- **Charts:** Chart.js 4.4.0

6.9 Study Buddy (`student/study_buddy.php`)

- **Features:** 1-to-1 buddy pairing, send/accept/decline requests by email, invite codes, unpair, block/unblock/report, privacy-safe progress comparison, nudge presets (wave, motivate, reminder, celebrate, challenge), full instant messenger
- **AJAX Endpoints:**
  - `action=send_message` → sends chat (rate-limited: 15/min)
  - `action=load_messages` → initial paginated load
  - `action=poll_messages` → new messages + typing/online status
  - `action=mark_read` → marks messages + nudges as read
  - `action=typing_start` / `typing_stop` → 3-second TTL indicators
  - `action=heartbeat` → online status ping
  - `action=delete_message` → sender-only deletion

- **Form Actions:** `send_request`, `accept_request`, `decline_request`, `unpair`, `block_buddy`, `unblock_buddy`, `report_buddy`

6.10 Buddy Messenger (`student/buddy_messenger.php`)

- **Included from** `study_buddy.php` when buddies are paired
- **Features:** Messenger-style chat UI, conversation bubbles, online indicator, typing indicator, reply-to threading, dropdown menu (report, block, unpair), report modal with reason categories

6.11 Global Search (`student/global_search.php`)

- **AJAX GET:** `?q=searchterm`
- **Searches:** tasks (title/description), notes (title/content), subjects (name/instructor)
- **Returns:** JSON array with type tags

6.12 Utility Endpoints

File	Method	Purpose
`student/dismiss_announcement.php`	AJAX POST	Inserts into `announcement_reads`
`student/dismiss_onboarding.php`	AJAX POST	Sets `onboarding_completed = 1`

---

7. Admin Modules

7.1 Admin Dashboard (`admin/admin_dashboard.php`)

- **Stats:** Total users, total tasks, completion rate, total study hours, semesters, subjects, new users this month, pending tasks

- **Tables:** Recent 5 users, top 5 active users by task count

7.2 Manage Users (`admin/manage_users.php`)

- **Features:** User list with search, filter by role, delete users (prevents last admin), change role (prevents last admin demotion), cleans up avatar files on delete
- **Actions:** `delete_user`, `change_role`

7.3 Announcements (`admin/announcements.php`)

- **Features:** Create/edit/delete, priority levels (Low/Normal/Important/Urgent), expiry dates, read count tracking per announcement
- **Actions:** `add`, `edit`, `delete`

7.4 System Settings (`admin/system_settings.php`)

- **Features:** Database record counts, admin password reset for any user, cleanup old completed tasks (min 30 days), cleanup old study sessions
- **Actions:** `reset_password`, `cleanup_completed`, `cleanup_sessions`

7.5 System Reports (`admin/system_reports.php`)

- **Features:** Summary cards, tasks by status/priority/type charts, monthly user registration trend, average tasks per user

7.6 Activity Log (`admin/activity_log.php`)

- **Features:** Filterable feed (All/Tasks/Study/Users), recent tasks, study sessions, registrations

7.7 User Details (`admin/user_details.php`)

- **URL:** `?id=X`
- **Features:** Full user profile, task stats, study sessions, semesters/subjects

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8. Auth Pages

Page	File	Key Features
Login	`auth/login.php`	CSRF, bcrypt verify, brute-force lockout, redirect support, role-based redirect
Register	`auth/register.php`	Validation (min 8 chars, uppercase + number), duplicate check, auto-redirect to login
Forgot Password	`auth/forgot_password.php`	Token via `random_bytes(32)`, SHA-256 hash stored, 1-hour expiry, demo mode shows link
Reset Password	`auth/reset_password.php`	Token validation, same password policy, marks token as used
Profile	`auth/profile.php`	Update name/email/course/year, change password, avatar upload (2MB, MIME validated)
Logout	`auth/logout.php`	Full session destroy + cookie clear

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9. Frontend JavaScript Systems

File: `assets/js/main.js` (1053 lines)

9.1 `StudifyToast`

Custom toast notification system. Methods: `.success()`, `.error()`, `.info()`, `.warning()`, `.show()`. Auto-removes after 4 seconds with exit animation.

9.2 `DarkMode`

Toggles `data-theme="dark"` on `<html>`, persists via `localStorage`. Updates sun/moon icon on toggle buttons.

9.3 `Sidebar`

Mobile hamburger toggle with overlay. Desktop collapse toggle persists via `localStorage`. Adds `sidebar-collapsed` class.

9.4 `ScrollAnimations`

`IntersectionObserver`-based `.fade-in-up` animation trigger for elements entering the viewport.

9.5 `LandingNav`

Adds `.scrolled` class to `<nav>` on scroll > 50px (landing page only).

9.6 `GlobalSearch`

- Opens via **Ctrl+K** / **Cmd+K** / `/`
- 300ms debounced AJAX search to `global_search.php`
- Arrow key navigation, Enter to select, ESC to close
- Groups results by type (Tasks / Notes / Subjects) with icons
- XSS-safe via DOM `textContent` → `innerHTML`

9.7 `KeyboardShortcuts`

Key	Action
Ctrl+K / **Cmd+K**	Open search
/	Open search
N	Navigate to tasks
P	Navigate to pomodoro
D	Navigate to dashboard
ESC	Close search

- Skips shortcuts in `<input>`, `<textarea>`, `<select>`, and `contenteditable` elements (Quill editor safe)

9.8 `FocusAmbiance`

- **Web Audio API** procedural sound generation — no external audio files
- **6 sounds:** Rain, Fire/Crackling, Ocean Waves, Birds/Chirps, Coffee Shop, Wind
- Individual volume sliders, simultaneous playback
- **5 presets:** Cozy (Rain+Fire), Nature (Birds+Waves), Study Café (Coffee+Rain), Storm (Rain+Wind), Zen (Birds+Wind)
- Auto-pause during breaks, auto-resume on focus
- Hooks: `.onFocusStart()`, `.onBreakStart()`, `.onPause()`

9.9 `StudifyConfirm`

Promise-based confirmation dialog system. Methods: `.show()`, `.action()`, `.delete()`, `.form()`, `.logout()`. Types: danger, warning, info, success. ESC / click-outside to cancel.

9.10 Helper Functions

Function	Purpose
`getCSRFToken()`	Reads from `<meta name="csrf-token">`
`toggleTaskStatus(id, base)`	AJAX POST to toggle task status
`deleteTask(id, base)`	AJAX DELETE with fade animation
`showConfirmDialog(...)`	Shorthand for `StudifyConfirm.show()`

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10. Layout & UI Architecture

Header (`includes/header.php` — 406 lines)

- **Meta tags:** `base-url`, `csrf-token`, `theme-color`, manifest link
- **Dependencies loaded:** Bootstrap 5.3 CSS, Font Awesome 6.4, FullCalendar 6.1.10 CSS

- **Sidebar:** Logo, role-based nav (student: 9 links + badge counts; admin: 6 links), user avatar footer
- **Topbar:** Hamburger toggle, page title, search button (Ctrl+K), dark mode toggle, notification dropdown (tasks + announcements + buddy messages), user avatar dropdown
- **Global Search Modal:** Full-screen overlay, keyboard navigation
- **Confirmation Dialog:** Reusable overlay for destructive actions
- **Onboarding Checklist:** 3-step guide (semester → subjects → task), dismissable
- **Flash Messages:** Session-based → toast conversion

Footer (`includes/footer.php`)

- Loads: Bootstrap 5.3 JS Bundle, FullCalendar 6.1.10 JS, Chart.js 4.4.0, `main.js` (cache-busted via `filemtime()`)

CSS (`assets/css/style.css`)

- Single monolithic stylesheet with `filemtime()` cache-busting
- **CSS custom properties** (`:root` vars) for all colors, spacing, shadows
- **Dark mode** via `[data-theme="dark"]` selectors
- **Layout:** Fixed sidebar (240px) + topbar + scrollable main
- **Responsive:** Sidebar overlay on mobile, collapse on desktop
- **Component styles:** Auth cards, stat cards, task cards (with priority borders + status badges), calendar, pomodoro ring, toast notifications, search modal, confirmation dialog, onboarding checklist, completed tasks section, ambiance grid, WYSIWYG editor

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11. PWA Support

File: `manifest.json`

```
{
  "name": "Studify",
  "short_name": "Studify",
  "description": "Academic Task Manager for Students",
  "start_url": "/Studify/",
  "display": "standalone",
  "background_color": "#F9FAFB",
```

```
  "theme_color": "#16A34A",
  "icons": ["192x192", "512x512"]
}
```

---

12. Security Summary

Feature	Implementation
CSRF Protection	Token per session via `random_bytes(32)`, validated on all POSTs, **rotated after use** (replay prevention)
Password Hashing	`password_hash(PASSWORD_BCRYPT)` / `password_verify()`
SQL Injection	Prepared statements (`bind_param`) throughout
XSS Prevention	`htmlspecialchars()` on all output; JS-side DOM `textContent` sanitization
Session Security	`session_regenerate_id(true)` on login, `use_strict_mode`, `cookie_httponly`, `cookie_samesite=Lax`, 2-hour idle timeout
Brute Force	Max 5 login attempts → 15-minute lockout
Input Sanitization	`trim` + `stripslashes` + `htmlspecialchars` via `cleanInput()`
File Upload	Extension whitelist, MIME validation, size limits (10MB/2MB), unique filenames
Authorization	`requireLogin()`, `requireAdmin()`, ownership checks on all data operations
Password Policy	Min 8 characters, requires uppercase + number
Chat Rate Limiting	Max 15 messages per minute
Token-based Reset	SHA-256 hashed, 1-hour expiry, single-use
Admin Safety	Prevents deleting/demoting last admin account

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13. Key Features at a Glance

Feature	Technology
🍅 **Pomodoro Timer**	SVG circular ring, vanilla JS, AJAX session save, task completion prompt
🎵 **Focus Ambiance**	Web Audio API procedural sound (6 environments, 5 presets)
📝 **Rich Text Notes**	Quill.js WYSIWYG editor with full toolbar, dark mode support
📅 **Calendar**	FullCalendar 6.1.10 with drag-to-reschedule
📊 **Study Analytics**	Chart.js 4.4 (bar, line, radar, doughnut charts)
👥 **Study Buddy**	1:1 pairing, real-time chat (polling), typing indicators, nudges, block/report
🔍 **Global Search**	Ctrl+K modal, debounced AJAX, keyboard navigation
🌙 **Dark Mode**	CSS custom properties, localStorage persistence
✅ **Task Management**	CRUD, status filtering, priority sorting, recurring, subtasks, separated completed view
📢 **Announcements**	Admin → student, priority levels, expiry, read tracking
🎓 **Onboarding**	3-step guided checklist for new users
📱 **PWA**	manifest.json, standalone display mode
⌨️ **Keyboard Shortcuts**	N/P/D/Ctrl+K navigation, Space for Pomodoro
🔒 **Security**	CSRF rotation, bcrypt, brute-force lockout, prepared statements, rate limiting

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This document covers all modules, functions, database tables, and features of the Studify system as of March 12, 2026.